

Trading and Exchanges: Market Microstructure for Practitioners

A Detailed, Section-by-Section Breakdown

Larry Harris — Oxford University Press, 2003 (sole edition). ISBN 0195144708.

Prepared as a study companion for the Trader Screener project. Organized to follow the book's thematic structure. Each section gives the core ideas, key definitions, worked intuition, and — where useful — a note on how it informs the screener build. Chapter topics are grouped by theme; consult the book's table of contents for exact chapter numbers.

How to read this document

The book is long (~640 pages, ~29 chapters). It is built around one question that recurs on nearly every page:

Why do you expect to profit from a trade? Who is on the other side, and why are they willing to lose to you?

Harris's thesis: trading is a **zero-sum game before transaction costs and a negative-sum game after them**. Profits come from other traders, so you must understand *who* trades, *why* they trade, and *how* the market structure distributes costs and information. The breakdown below walks every major theme that answers that question.

PART I — INTRODUCTION: WHAT TRADING IS

1. The purpose of markets

Markets perform two economic functions: - **Price discovery** — aggregating everyone's information and demand into a single clearing price. - **Liquidity / risk transfer** — letting people exchange cash for assets (and risk) whenever they need to.

A *good* market does both cheaply, fairly, and reliably. Microstructure is the study of *how the rules and mechanics of a market* affect those outcomes.

Key distinction: *investing* is deciding **what** to hold; *trading* is the act of **exchanging** — and how well you trade (cost, timing, execution) is a separate skill from what you choose to hold.

2. Instruments traded

Harris surveys the instrument universe and *why each exists*: - **Equities** — ownership; claim on residual cash flows. - **Debt / fixed income** — lending; claim on fixed payments. - **Currencies (forex)** — medium of exchange across economies. - **Commodities** — physical goods; consumed or used as inputs. -

Derivatives — forwards, futures, options, swaps — contracts whose value derives from an underlying. They exist to **transfer specific risks** (price, rate, FX) without trading the underlying itself.

Each instrument exists because someone needs to move value through **time** (investors/borrowers), **place** (currencies), or **risk-state** (hedgers via derivatives).

3. The trading industry — the cast

The ecosystem: **buy side** (those who use liquidity — investors, hedgers, etc.), **sell side** (those who supply it — dealers, brokers), and the **support infrastructure** (exchanges, clearinghouses, regulators, data vendors). Understanding who profits from whom starts with knowing each role.

***Screener relevance:** the screener serves a buy-side user picking what to hold/trade. The instrument list here is the menu for shortlist §1.*

PART II — ORDERS AND MARKET STRUCTURES

4. Orders — the atoms of trading

An order is an instruction to trade with specific properties. The two foundational types embody the master trade-off of all trading: **immediacy vs. price**.

Market orders

- Execute **immediately** at the best available price.
- **Demand liquidity** — you take what's offered.
- **Cost:** you pay the bid/ask spread and suffer **market impact** (your order pushes the price).
- Use when **certainty of execution** matters more than price.

Limit orders

- Execute only at your specified price **or better**; may not execute at all.
- **Supply liquidity** — you post a price others can hit.

- **Two risks:** 1. **Non-execution risk** — the market moves away and you never get filled (you miss the trade). 2. **Adverse-selection risk** — you get filled mainly when you're *wrong* (someone with better information hit your stale quote).
- Use when **price** matters more than immediacy and you can be patient.

Stop orders

- Dormant until price reaches a **trigger**, then convert to a market (or limit) order.
- Used for **loss-cutting** (sell-stops below your entry) and **breakout entry** (buy-stops above resistance).
- **Danger:** stops demand liquidity exactly when it's scarcest (during a fast move), so they often fill at poor prices ("slippage"). A stop is not a guaranteed exit price.

Order properties

- **Validity:** day, good-till-cancelled (GTC), immediate-or-cancel (IOC).
- **Quantity:** all-or-none (AON), minimum fill, **hidden / iceberg** (display only part of size).
- **Price/execution instructions:** market, limit, market-on-close, etc.

5. Market structures — where orders meet

The same instrument can trade in different structures, each distributing cost and information differently.

Quote-driven (dealer) markets

- **Dealers** post continuous bid and ask quotes; you trade *against the dealer*.
- The dealer provides **immediacy** and earns the **spread** as compensation.
- Classic example: traditional NASDAQ, bond markets, FX.

Order-driven (auction) markets

- Buyers and sellers trade **with each other** through a central **limit order book**; no obligatory dealer.
- Two timing models:
- **Continuous markets** — a trade happens any moment two orders cross.
- **Call (batch) auctions** — orders accumulate and clear together at a single price at a scheduled time (opening/closing auctions).

Brokered markets

- **Brokers actively search** for counterparties. Used for large, illiquid, or unique instruments (big equity blocks, real estate) where posting publicly would move the price.

Hybrid markets

- Most modern venues combine these (e.g., an order book plus designated market makers).

Order precedence rules (how the book decides who trades)

1. **Price priority** — best price always trades first (highest bid, lowest ask).
2. **Secondary precedence** among equal prices — typically **time** (first to post), sometimes **display** (visible before hidden) and **size**.

Knowing precedence tells you whether *your* limit order will realistically get filled, or whether you're stuck behind a long queue.

Screeners relevance: order mechanics define what "liquidity" and "spread" mean. A thin order book = wide spread + high impact = signals on illiquid names are hard to act on. Feeds shortlist §2c.

PART III — THE PLAYERS: WHO TRADES AND WHY

This is the heart of the book — Harris's famous **taxonomy of traders**. Split first by *motive*.

6. Utilitarian traders — trade for reasons other than trading profit

They expect to *lose* a little on trading because trading serves another goal. They are the **ultimate source of profit** for everyone else. - **Investors** — move money through time (save now to spend later). - **Borrowers** — raise money now (issue debt/equity). - **Hedgers** — offload unwanted real-world risk (airline hedges jet fuel; farmer hedges crop price). - **Asset exchangers** — need the asset itself (buy currency to pay a foreign invoice). - **Gamblers** — trade for the entertainment/utility of the bet. - **Cross-subsidizers & forced traders** — index funds rebalancing, traders forced by tax, margin, or mandate.

Because these traders don't trade *to win*, they reliably hand profits to informed traders and liquidity suppliers.

7. Profit-motivated traders — trade specifically to make money

Informed traders (make prices *informative*)

- **Value traders** — independently estimate fundamental value; buy below, sell above. The **deepest, most stabilizing** liquidity source; they step in when prices dislocate.
- **News traders** — act on new public information faster than others.

- **Arbitrageurs** — exploit price discrepancies between related instruments; enforce the **law of one price**. Provide liquidity in one instrument by hedging in another.
- **Information-oriented technical traders** — infer information from price/volume patterns (legitimate when patterns reflect real order flow).

Dealers / market makers (supply liquidity)

- Profit from the **spread**, not direction. Continuously quote both sides, manage inventory, and aim to be flat over time.

Order anticipators (parasitic — profit from *others' orders*, add no information)

- **Front runners** — trade ahead of a known large order (illegal when based on client info).
- **Sentiment-oriented technical traders** — try to detect and ride waves of uninformed trading.
- **Squeezers** — exploit traders who *must* trade (short squeezes, corners).

Bluffers / manipulators

- Use deceptive orders/trades to mislead others about value or demand, then profit from the reaction. Illegal (manipulation).

8. Brokers and the agency problem

Brokers trade **as agents** for clients. Harris details their **conflicts of interest**: - **Agency vs. principal** — is the broker filling you from the market or from their own book? - **Soft dollars** — paying for research with client commissions. - **Payment for order flow** — being paid to route your order to a particular venue (may not get you the best price). - **Best execution** obligations exist precisely because incentives can diverge from the client's.

9. Why people trade (and the brutal arithmetic)

Since trading is zero-sum before costs, **for every winner there is a loser**. Profit-motivated traders win by trading against **utilitarian** and **uninformed** traders. Harris's recurring warning:

If you can't identify who is losing to you and why they keep coming back, you are the one losing.

Screeners relevance: This taxonomy is the justification for the smart-money moat. **Corporate insiders and well-connected politicians are textbook informed traders**. A screener that surfaces their activity is an informed-trader-detection tool — Harris explains exactly why piggybacking informed flow is an edge. Feeds shortlist §2e.

PART IV — LIQUIDITY, THE SPREAD, AND VOLATILITY

10. Why the bid/ask spread exists — its three components

The spread is **not** dealer greed; it compensates three genuine costs: 1. **Order-processing costs** — the operational cost of making markets (technology, clearing, settlement, the dealer's time/capital). 2. **Inventory-holding costs** — when a dealer takes the other side of your trade, they hold a position exposed to price risk until they can offload it. They charge for bearing that risk. 3. **Adverse-selection costs** — the decisive one. Dealers systematically **lose to informed traders**, who only trade when they know the price is wrong. To survive, dealers **widen the spread** and recover those losses from **uninformed** traders.

Implications: - Spreads **widen** with information asymmetry, uncertainty, and volatility. - Spreads **narrow** with competition and high uninformed (liquidity) volume. - A persistently **wide spread** signals thin competition and/or high adverse-selection risk — a hazard for traders.

11. Adverse selection and the winner's curse

When your limit order fills **instantly and completely**, ask *why someone was so eager to take your price*. Often: they know something you don't. Liquidity suppliers face a constant **winner's curse** — the market accommodates you most readily exactly when you're on the wrong side. Surviving as a liquidity supplier means pricing this risk into the spread.

12. Liquidity — Harris's four dimensions

Liquidity is **not a single number**. It has four distinct dimensions: 1. **Immediacy** — how quickly you can trade a given size. 2. **Width** — the bid/ask spread (the cost of immediacy). 3. **Depth** — how much size you can trade *without moving the price*. 4. **Resiliency** — how quickly the price recovers after a liquidity-demanding trade.

A truly liquid instrument scores well on **all four**. One number (e.g., share volume) can hide poor depth or resiliency.

13. Volatility — two kinds

- **Fundamental volatility** — price movement from *real information* arriving (earnings, macro data). Legitimate and unavoidable.

- **Transitory volatility** — noise from microstructure: **bid/ask bounce** (price flickering between bid and ask), temporary order imbalances, liquidity demand. Not "real" value change.

Distinguishing the two matters: transitory volatility creates the illusion of risk/opportunity that isn't fundamentally there.

Screeners relevance: Directly specs shortlist §2b (volatility: ATR, realized vol, beta) and §2c (liquidity: ADV, dollar volume, spread). Harris's four-dimension model is the reason **dollar volume** (a depth proxy) beats raw share count as a tradeability filter.

PART V — TRANSACTION COSTS AND PERFORMANCE MEASUREMENT

14. The full cost of trading

Costs are **explicit + implicit**: - **Explicit** — commissions, exchange fees, taxes. Visible, usually small. - **Implicit** — the larger, hidden costs: - **Spread cost** — half the spread per side, paid every round trip. - **Market impact** — your own order moving the price against you (worse for large orders). - **Opportunity / delay cost** — the price drifting while you wait to execute, plus the cost of trades you **fail to complete**.

For large or impatient traders, implicit costs **dwarf** commissions.

15. Implementation shortfall — the honest cost metric

Harris champions **implementation shortfall**: the difference between - the **paper portfolio** (value if you'd executed instantly at the price when you *decided* to trade), and - the **actual portfolio** (what you really achieved, including unfilled trades).

It captures **delay cost and missed-trade cost** that simpler measures hide. This is the gold-standard way to measure whether you're trading well.

16. VWAP and other benchmarks

VWAP (volume-weighted average price) is a popular execution benchmark but has flaws: it's **gameable** (trade with the volume to match it), and it **ignores opportunity cost** (beating VWAP while the stock runs away is a hollow victory). Use it with awareness of its limits.

17. Why this is the most controllable edge

Since trading is negative-sum after costs, **minimizing transaction costs is the single most reliable, controllable edge** for most traders. You can't control the market; you *can* control how much you bleed entering and exiting.

Screener relevance: *This is the case for liquidity/cost columns (ADV, dollar volume, spread) as first-class filters — they predict what a trade will cost. A brilliant signal on an illiquid name is worthless if impact eats the edge before you're filled.*

PART VI — MARKET EFFICIENCY, STRUCTURE, AND PATHOLOGIES

18. Informative prices and the Grossman–Stiglitz paradox

Prices are informative **because informed traders trade** on their information. But informed traders need **uninformed (liquidity) traders to hide behind** — they profit by disguising their informed orders among noise. The paradox: - If markets were *perfectly* efficient, no one could profit from information, - so no one would pay to gather information, - so prices would **stop** reflecting information.

Therefore markets must be **slightly inefficient** to reward information gathering. A healthy market needs a **mix** of informed and uninformed participants.

19. Market efficiency in practice

Efficient markets reflect available information; **microstructure frictions** (spreads, latency, order-handling rules) determine *how fast and how fully* prices incorporate news. "Efficiency" is a spectrum set by structure, not a binary.

20. Order-driven market mechanics in depth

- **Call auctions** — concentrate liquidity at a point in time; good for opening/closing and illiquid names; produce a single fair clearing price.
- **Continuous order books** — provide immediacy but fragment liquidity across time.
- **Trade-offs:** continuous trading offers immediacy at the cost of thinner instantaneous liquidity; call auctions offer depth at the cost of waiting.

21. Block trading and the upstairs market

Large orders avoid the public book to escape **market impact**: - **Upstairs market** — brokers negotiate large blocks privately, then print them. - **Hidden / iceberg orders** — show only a fraction of true size. - **Dark pools** (post-2003 evolution) extend this idea — trade size without revealing intent.

22. Manipulation, bubbles, and crashes

- **Manipulation tactics** — Harris catalogs pump-and-dump, cornering, squeezes, spoofing-type deception, and marking the close.
- **Bubbles and crashes** — prices detach from value when feedback trading dominates; **crashes cluster** because liquidity *evaporates exactly when everyone demands it at once* (a self-reinforcing liquidity spiral).

23. Insider trading and regulation

- **Why it's restricted**: insider trading is the ultimate **adverse selection**. If uninformed traders know insiders will pick them off, they leave — destroying liquidity for everyone.
- Regulation exists to **protect liquidity and fairness**, keeping uninformed participants willing to trade.

***Screeners relevance:** Harris's insider-trading discussion is the **academic backbone** for treating legal, disclosed insider/congressional filings as informed-trader signal. He explains both why such information is powerful **and** why markets regulate the illegal form of it.*

PART VII — PRACTICAL LESSONS (THE TAKEAWAYS TRADERS QUOTE)

1. **Know why you trade.** Have a concrete, defensible reason you expect to profit, and name who you profit *from*. No answer → you are the liquidity others feed on.
2. **"If you don't know who the sucker is, it's you."** (The book's most-quoted line, paraphrased.)
3. **Minimize transaction costs.** The most reliable, controllable edge — costs compound against you every round trip.
4. **Match order type to need.** Limit orders when patient (supply liquidity, earn spread, accept non-execution risk); market orders when you need immediacy (demand liquidity, pay spread).
5. **Liquidity is multi-dimensional.** Don't confuse a tight spread with depth, or volume with resiliency. Check all four dimensions.
6. **Respect adverse selection.** Easy, instant fills are suspicious. The market accommodates you most when you're wrong.
7. **Information edges decay at different rates.** Value-based edges are durable; speed/anticipation edges get competed away fast.

8. **Distinguish fundamental from transitory volatility.** Don't trade noise as if it were signal.

HOW THIS BOOK SHAPES THE TRADER SCREENER

Harris concept	Build implication for the screener
Trader taxonomy / informed traders	The smart-money moat (congress + insider + committee data) is literally informed-trader detection — make it a first-class, dedicated view/filter.
Liquidity's four dimensions	Liquidity columns must go beyond share volume — prefer dollar volume (depth proxy); flag wide spreads.
Transaction costs / impact	ADV, dollar volume, and spread are tradeability gates — a signal is only actionable if the name can be traded cheaply.
Spread = adverse selection	Wide-spread / thin names carry hidden risk; surface this as a warning, not just a number.
Fundamental vs transitory volatility	Volatility columns (ATR, realized vol, beta) are for sizing and stops , not standalone signals.
Execution is its own discipline	Don't try to be an order-routing/execution venue — the screener's edge is research and screening (consistent with the project's architecture).
"Know who's on the other side"	Every screener filter should help answer <i>who is informed here, and can I ride them?</i>

End of detailed breakdown. Pair this with the condensed [summary](#) for quick reference, and use the vocabulary here to fill the [Screener Shortlist](#).